SOMIN IVY YOON

217-778-7630 |Email: ivyyoon2@illinois.edu | www.linkedin.com/in/ivyyoon/

EDUCATION

University of Illinois at Urbana-Champaign Expected May 2028
Bachelor of Science in Mechanical Engineering | Minor in Electrical Engineering GPA: 3.70/4.00

TECHNICAL SKILLS

CAD & Simulation: NX, PTC Creo, Inventor, AutoCAD, Fusion360, ANSYS, SolidWorks
Programming: Java, MATLAB, Python, C/C++, JavaScript, VBA
Fabrication: CNC/Manual Machining, TIG/ARC/MIG Welding, Soldering, 3D Printing
Embedded Systems: PLC, Arduino, Raspberry Pi, ESP32, STM32, ROS

WORK EXPERIENCE

Littelfuse, Inc. Champaign, IL
Engineering Intern May 2026 – Aug 2026

- Devised universal IEC/UL testing and X-ray fixtures for fuses in Autodesk Inventor, reducing setup time by 80% (60s to 12s)
- Conducted stress analysis of laser labeling fixtures under 3+ loads, evaluating deformation across 10+ design/material options
- Developed Excel VBA automation for the laser labeling process, increasing throughput to 100 engineering samples in 30 min
- Built a load-cell controlled pneumatic dispensing system on an Arduino Opta PLC, cutting cycle time by 37% through FEA and testing

Yuil Trading Corporation Seoul, Korea
Manufacturing Intern May 2025 – Aug 2025

- Designed a 2D AutoCAD support fixture to prevent oil backflow in a hydraulic press to improve operating table reliability
- Assembled housings and electronics for 250+ hospital suction units, testing vacuum level, free air flow, and leakage to ISO 10079
- Linked pressure transmitters to Excel via DAQ to calculate flow rates for 30+ suction units per day, flagging units below 20 L/min

PROJECTS

Wheel Speed Sensor PCB *Eco Illini Supermileage* Aug 2026 – Present

- Designing a hub mount and low friction frame housing that hold the magnet on the sensor axis at a 1.5 mm air gap
- Designed a KiCAD PCB around an STM32G0B1 that counts quadrature pulses on a hardware timer to measure rear wheel speed
- Isolated the CAN bus with an ISO1044 transceiver, sending data and 12 V power between boards over RJ45 and Ethernet cable
- Calculated a one-pulse-per-rev Hall switch updates only once per second at 3.6 mph, leading to a 14-bit angle encoder (MA732)

Shock Tuning Data Acquisition System *Off-Road Illini Baja SAE* May 2026 – Present

- Leading mechanical, firmware, and data subteams tuning shocks against bottoming out with measured data instead of driver feel
- Designing 3D printed split clamp mounts fixing linear potentiometers to shocks, offset 5 mm to protect sensor travel at compression
- Developed an ESP32-S3 logger and C pipeline turning travel data from four shocks into velocity histograms for damper tuning
- Selected a 12-bit SPI ADC (MCP3208) at 200 to 500 Hz after noise analysis showed 10-bit resolution masked slow damper motion

Adaptive Gravel E-Trike *American Society of Mechanical Engineers* Sep 2024 – Present

- TIG-welded 6061 aluminum rods and formed sheet metal to integrate the designed footplate with the trike frame in clearance fit
- Evaluated bending stiffness and structural integrity under 10+ static load cases, measuring vertical deflection and deformation
- Earned 1st place in Most Industry Impact and 2nd place in Outstanding RSO among 200+ exhibits at UIUC Engineering Open House

AI Translator Mask *American Society of Mechanical Engineers* Dec 2024 – Apr 2025

- Led a team of 10 to build a 3D printed Raspberry Pi mask with a speaker and TFT display that translates and captions speech live
- Reduced translation latency by 24% by restructuring the processing workflow and optimizing Google Cloud API call sequencing
- Earned 1st place in Visionary Impact among 200+ exhibits at UIUC Engineering Open House 2025

EXTRACURRICULAR ACTIVITIES & LEADERSHIP EXPERIENCE

American Society of Mechanical Engineers Champaign, IL
Director of Special Projects Aug 2025 – May 2026

- Recruited 70+ members by presenting for 15 minutes in core ME classes and facilitated weekly meetings of 30+ committee members
- Oversaw 6 engineering projects ranging from robotics to aeronautics, coordinating budgets and technical milestones